

Query Relation: has\_profession

Interpretation: Actors graduated from the same school are likely to share the same profession.

|        |                            |                |                               |                |                              |                 |                      |
|--------|----------------------------|----------------|-------------------------------|----------------|------------------------------|-----------------|----------------------|
| True:  | [Actor]<br>John Rhys-Davis | ←has_graduates | [Thing]<br>RADA               | has_graduates→ | [Actor]<br>Micheal Gambon    | has_profession→ | [Thing]<br>Actor     |
| False: | [Actor]<br>Peter Ustinov   | ←has_graduates | [Thing]<br>Westminster School | has_graduates→ | [Author]<br>Christopher Wren | has_profession→ | [Thing]<br>Scientist |

Query Relation: has\_genre

Interpretation: Films focusing on the same subject are likely to share the same genre. However, films depicting the same profession can have different genres.

|        |                              |           |                          |           |                         |            |                             |
|--------|------------------------------|-----------|--------------------------|-----------|-------------------------|------------|-----------------------------|
| True:  | [Topic]<br>Life Is Beautiful | ←has_film | [Subject]<br>World War 2 | has_film→ | [Topic]<br>Valkyrie     | has_genre→ | [Genre]<br>War Film         |
| False: | [Topic]<br>Air Force One     | ←has_film | [Profession]<br>Aviation | has_film→ | [Topic]<br>Team America | has_genre→ | [Genre]<br>Political Satire |

Query Relation: nominee\_work

Interpretation: The best actors are likely to perform great in all movies. The same company may not always produce award winning works.

|        |                                 |               |                           |                |                         |                |                            |
|--------|---------------------------------|---------------|---------------------------|----------------|-------------------------|----------------|----------------------------|
| True:  | [Thing]<br>Academy Actor        | nominee_work→ | [Award Topic]<br>Network  | ←nominated_for | [Actor]<br>Sidney Lumet | nominated_for→ | [Award Item]<br>Orient Exp |
| False: | [Award Category]<br>Emmy Comedy | nominee_work→ | [Tv Program]<br>Entourage | ←nominated_for | [Employer]<br>HBO       | nominated_for→ | [Tv Program]<br>Gia        |

Table 4: Correctly predicted examples by the proposed model on test data of FB15k-237. Despite the fact that each pair of true and false examples share the same sequence of relations, the uses of entity types at the right level of abstraction (shown in bold) help distinguish between them. Abbreviated entity and relation names are shown for clarity.

|                  | WN18RR       | FB15k-237    |
|------------------|--------------|--------------|
| TuckER           | 38.16        | 40.12        |
| ComplEx-N3       | 40.89        | 40.35        |
| APR <sup>D</sup> | <b>84.91</b> | <b>57.35</b> |

Table 5: MAP% comparison to two SOTA embedding methods.

We also examined the distribution of attention weight at different levels of type hierarchies. Figure 4 shows that the model leveraged all levels for representing various entities. More specifically, level 1, 2, and 3 are most commonly emphasized for entities in WN18RR; level 1,2, and 6 are most often used for entities in FB15k-237. The emphasis on lower levels is expected since using the most specific level is more beneficial than using the most abstract level with evidence in Table 3. However, the diversity of levels used by the model proves that not all entities should use the most specific level: different entities require different levels of abstraction.

Finally, we visualized examples of paths from prediction along with the top weighted types the model selected for entities. In Table 4, we are able to see examples that correspond strongly with human intuition. This qualitative result again verifies that the model learns a new class of rules (path patterns) that is more specific yet still generalizable.

## Comparison with Knowledge Graph Embedding Methods

As a final point of comparison, we validated our best model *APR<sup>D</sup>* against two state-of-the-art KGE methods<sup>7</sup>: *TuckER* (?) and *ComplEx-N3* (?). As shown in Table 5, our model

<sup>7</sup>We followed (?) to evaluate KGE methods on the fact prediction task. Test triples with the same relation are ranked.

performs significantly better on both datasets. In fact, all neural-based path ranking methods (*APR* and *Path-RNN*) outperform the KGE methods. One possible reason that path ranking methods outperform KGE is that KGE methods require both the source  $e_i$  and target entities  $e_k$  in the missing triple  $(e_i, r_j, e_k)$  to be known, thus failing to perform well when test entities are not present in the training set. However, only 17.6039% of FB15k-237’s test data consists of new entities, and only 0.9412% in WN18RR. As a result, new entities do not alone account for the difference in performance. This comparison additionally affirms the improved state-of-the-art performance in fact prediction achieved by our method.

## Conclusion

This work addresses the problem of knowledge base completion. We introduced Attentive Path Ranking, a novel class of generalizable path patterns leveraging type hierarchies of entities, and developed an attention-based RNN model to discover the new path patterns from data. Our approach results in statistically significant improvement over state-of-the-art path-ranking based methods and knowledge graph embedding methods on two benchmark datasets WN18RR and FB15k-237. Quantitative and qualitative analyses of the discovered path patterns provided insights into how APR achieves a balance between generalization and discrimination.

## Acknowledgments

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